Transformers Vs CNN For Audio Deepfake Detection: A Comparative Study and Cross-Dataset Evaluation

Abstract

Recently, deep learning has witnessed a paradigm shift with the adaptation of Transformer architectures form natural language processing (NLP) to computer vision tasks. High performing models such as the Vision Transformer (VIT) and Swin Transformer have demonstrated state of the art capabilities; however, their reliance of massive pre-training infrastructure and millions of images often limits their localized adoption.

Audio deepfake detection models often achieve high performance on the datasets used for their development, but their ability to generalize to unseen datasets and speech-generation methods remains a significant challenge. This work presents a rigorous comparative evaluation of conventional Convolutional Neural Networks (CNNs)- specifically ResNet18 and Resnet50- against modern vision-centric Transformer frameworks, including VIT, Swin Transformer, a custom CNN-Transformer hybrid, and Data-efficient Image Transformer (DEIT). Evaluating these models on the ASVspoof2019 Logical Access (LA) dataset consisting of 121,461 audio samples, we frame the task as a binary classification problem to distinguish between authentic (Bonafide) and synthesized (Spoof) acoustic signals. Performance was assessed on the ASVspoof2019 test split and subsequently examined across three external evaluation datasets: ASVspoof2021 Deepfake (DF), In-the-Wild and the LJSpeech-MelGAN subset of WaveFake.

Following fine-tuning, the Swin Transformer with eight attention heads achieved the strongest performance on the ASVspoof2019 LA test set, attaining an accuracy of 91.95%, a ROC-AUC of 0.9844, and an equal error rate (EER) of 5.86%. However, cross-dataset evaluation revealed substantial degradation in detection performance and changes in the relative ranking of architectures. On ASVspoof2021 DF, the CNN–Transformer hybrid achieved the lowest EER of 20.70%, with a ROC-AUC of 0.8792. On the In-the-Wild dataset, ResNet18 and the CNN–Transformer achieved the lowest EER of 47.36%, indicating limited discrimination between bona-fide and spoofed recordings. On the LJSpeech–MelGAN subset of WaveFake, the eight-head Swin Transformer achieved the best recorded results among the evaluated architectures, with a ROC-AUC of 0.7365 and an EER of 31.63%, despite an accuracy of approximately 50% at the frozen decision threshold.

These findings demonstrate a substantial gap between in-domain performance and cross-dataset generalization in audio deepfake detection. Although Transformer-based and hybrid architectures achieved strong results on ASVspoof2019 LA, their performance varied considerably across external datasets representing different synthesis methods and recording conditions. The results emphasize the need for multi-dataset evaluation when assessing the robustness of audio deepfake detectors and indicate that high validation accuracy alone is insufficient to establish reliable performance on unseen audio distributions.

1. Introduction and Architectural Paradigms:

The rapid proliferation of sophisticated generative adversarial networks (GANs) and neural speech synthesis algorithms has made the generation of highly realistic audio deepfakes accessible and widespread. These synthetic audio vectors present severe security threats, ranging from biometric identity theft and voice-clone bank fraud to the spreading of automated political disinformation. Consequently, developing robust, highly generalizable automated spoofing counters (ASVs) has become a critical focal point in contemporary multimedia security research.

Historically, the domain of audio deepfake detection has been dominated by Convolutional Neural Networks (CNNs). CNNs possess an inherent structural advantage known as spatial inductive bias. When raw audio waveforms are translated into two-dimensional time frequency representations- such as Short-Time Fourier Transforms (STFT) or Mel-Spectrograms- standard convolutional filters seamlessly treat them as images. Through localized receptive fields and shared weight matrices, CNNs are exceptionally efficient at capturing low-level acoustic anomalies, transient phase shift, and localized high-frequency smoothing patterns that characterize synthetic speech generation.

Conversely, the emergence of the Vision Transformer (ViT) has introduced an alternative processing framework. By discarding localized convolutional strides and instead flattening spectrogram grids into linear patches, tokens are processed via multi-head self-attention mechanisms. This enables the model to look past localized features and evaluate global, long-range temporal-frequency dependencies across the entire duration of an acoustic sample.

However, this global express-capacity comes at a steep training cost. Transformers lack a CNN's built-in spatial assumptions, meaning they require significantly longer optimization paths or massive datasets to discover sharp minimums on a loss surface. When prematurely restricted by tight early stopping windows during optimization, transformers frequently underperform, failing to catch simpler baseline networks.

Scope of Contribution:

As the sole investigator of this project, I designed, implemented, and optimized the entire machine learning pipeline from raw audio ingestion to final model distillation. The specific contributions of my independent work are as follow:

- **End-to-End Pipeline Engineering:** I built a highly optimized data processing pipeline that decouples feature extraction from backpropagation by pre-computing spectrogram tensors to ram and pre caching that reduced the compute time significantly, eliminating CPU storage bottlenecks. I also integrated a class-weighted random sampler to neutralize severe dataset imbalances.
- **Rigorous Empirical Benchmarking:** I independently managed the configuration, tracking, and training of six distinct deep learning frameworks: two pure CNN variants (ResNet18, ResNet50), a global projection transformer (ViT), a localized hierarchical window transformer (Swin), a cross-attention CNN-Transformer hybrid, and a teacher-student distilled framework (DeiT).
- **Discovery of Optimization Thresholds:** I uncovered critical limitations in standard early stopping routines. By designing a multi-stage empirical study, I demonstrated how shifting from constrained training to an unconstrained 100-epoch baseline alters the

global leaderboard—culminating in the discovery that the Swin Transformer's shifted window approach outperforms traditional architectures when given the necessary training runway.

2. Methodology and Model Selections

In this section, I will detail the specific variations of the network I implemented, customized, and adapted to parse single-channel audio spectrogram topologies.

2.1 Benchmark CNN Baseline: ResNet18 vs ResNet50 Capacity

To establish rock-solid baselines, I selected two Residual Network architectures differing significantly in depth and parameter capacity:

- **ResNet18:** A shallow, 18-layer deep residual architecture. Its smaller capacity allows it to optimize rapidly and capture local spectrogram anomalies cleanly without overfitting.
- **ResNet50:** A deeper, 50-layer architecture utilizing bottleneck blocks. While it risks overfitting on small acoustic datasets due to its massive parameter size, its deeper feature hierarchy allows it to extract abstract, high-dimensional acoustic representations if given an unconstrained training horizon.

Both architectures were modified at their initial layer to accept single-channel (1-channel) grayscale audio spectrogram tensors instead of standard 3-channel RGB images (***adapted from K.He [4]***)

2.2 Hierarchical Shifted Windows: The Swin Transformer Approach

The Swin Transformer bridges the gap between CNN efficiency and Transformer expressiveness. Instead of computing global attention across the entire spectrogram (which scales quadratically with image size), Swin computes attention strictly within localized, non-overlapping 2D windows. To preserve cross- window communication, it introduces a **shifted windowing mechanism** between consecutive layers. This builds a hierarchical representation, mimicking the feature- pooling nature of CNNs while maintaining the dynamic weight allocation of self-attention.

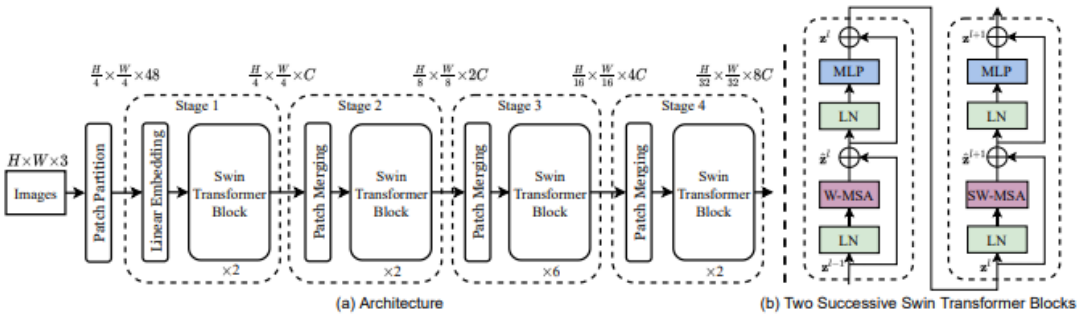


Fig 1: The hierarchical architecture of the Swin Transformer alongside its core internal block dynamics (adapted from Liu et al. [3]). As detailed in (a), the framework downsamples time-frequency dimensions across four discrete stages via Patch Merging to simulate a CNN's feature hierarchy. Part (b) outlines two successive blocks: the first applies standard Window-based Multi-Head Self-Attention (W-MSA) within isolated

boundaries, while the second introduces a Shifted Window approach (SW-MSA) to establish cross-window connectivity across the acoustic spectrogram.

2.3 Vision Transformers (VITs): Capturing Global Acoustic Context

Conversely, the emergence of the Vision Transformer (ViT) has introduced an alternative processing framework that completely discards localized convolutional strides. By flattening spectrogram grids into sequential, non-overlapping linear patches, tokens are processed via multi-head self-attention mechanisms. This allows every single patch to interact with every other patch across the entire time-frequency domain simultaneously. This enables the model to look past localized features and evaluate global, long-range temporal-frequency dependencies across the entire duration of an acoustic sample. However, because transformers lack built-in spatial assumptions, they have a completely flat inductive bias. They require significantly longer optimization paths to find sharp minimums on a loss surface.

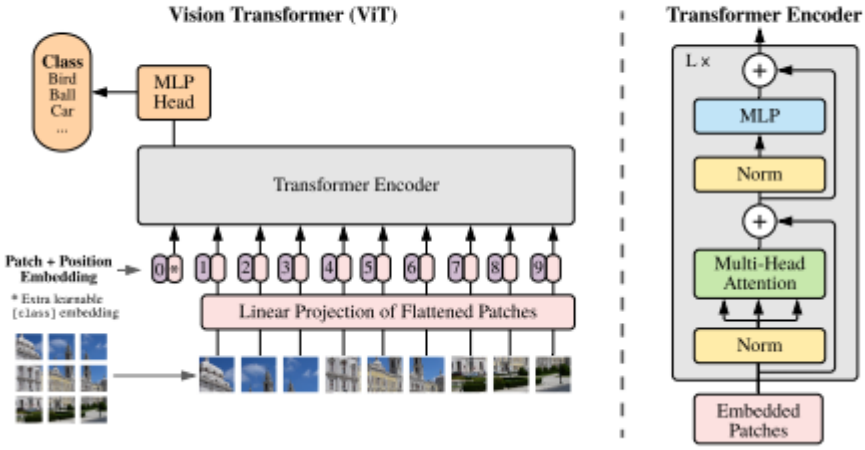


Fig 2: ***The foundational Vision Transformer (ViT) framework (adapted from Dosovitskiy et al. [2]). In my implementation, the standard 3-channel image input is substituted with a single-channel pre-computed audio spectrogram, which is subsequently flattened into sequential 2D time-frequency patch tokens before undergoing global multi-head self-attention.***

2.4 Cross- Attention Hybrids: Integrating CNN Front-Ends with Transformer Tails

I implemented a custom **CNN-Transformer Hybrid** designed to harvest the benefits of both worlds. The front-end consists of standard convolutional blocks that downsample the raw audio spectrogram and extract highly robust localized features. These feature maps are then flattened and treated as sequential tokens by a downstream Transformer encoder tail. This setup uses the CNN to enforce a strong early inductive bias, allowing the transformer layers to focus purely on modelling long-range temporal relationships among highly polished features.

2.5 Data-Efficient Image Transformers (DeiT): Knowledge Distillation Framework

The Data-Efficient image Transformer (DeiT) explicitly targets the data-hungry nature of standard transformers by introducing knowledge distillation. Alongside the standard classification token, DeiT introduces a specialized **distillation token** into its sequence. During training, this token interacts with the patch tokens via self-attention and is forced to minimize a distillation loss relative to the output of a strong, frozen teacher network of **Resnet 50** as it was the most efficient and highest accuracy model among the CNN models**.** This allows the transformer to directly inherit the localized inductive biases of a CNN without altering its core attention blocks. I tracked two distinct variants of this framework: **DeiT-Hard** (which forces the student to match the hard argmax label of the teacher) and **DeiT-Soft** (which uses KL-divergence to match the teacher's smooth logit probability curves).

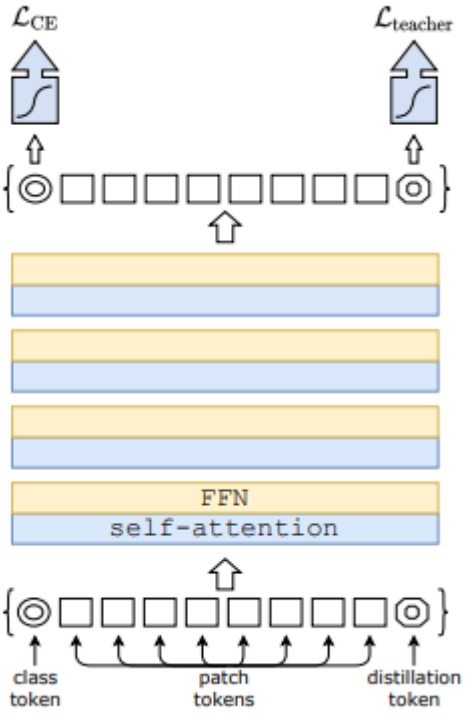


Fig 3: ***The DeiT teacher-student knowledge distillation pipeline framework featuring the specialized distillation token. (Reprinted from Touvron et al. [1]).***

Experimental Setup and Optimization Framework

3.1 Dataset Profiling and Spectrogram Feature Extraction

For all experiments, I utilized the standardized **ASVspoof2019 Logical Access (LA)** dataset ***(adapted by Massimiliano Todisco [6])***, which contains **121,461** audio files divided into training, validation, and testing partitions to evaluate Bonafide versus spoofed speech

To eliminate massive input/output hardware bottlenecks, I designed a pre-computing pipeline. Rather than executing expensive on the fly short Time Fourier Transforms (STFT) inside the dataloader on every epoch, I processed the raw waveforms into static, optimized 2D spectrogram matrices and saved them directly to disk as tensor arrays beforehand. During training, the dataloader performs a raw disk read of these pre-computed tensors, isolating

model performance from CPU calculation latency. Furthermore, because spoofing datasets are heavily skewed due to the sheer volume of synthetic variants generated per audio source, I heavily skewed due to the sheer volume of synthetic variants generated per audio source, I integrated a **WeightedRandomSampler** into the Pytorch pipeline. By adjusting batch drawing probabilities based on reciprocal of class frequencies, I ensured every mini-batch maintained a strict, mathematically balanced **50/50** distribution between real and fake samples.

3.2 Hardware and Hyperparameter Settings

All frameworks were compiled and executed within a dedicated NVDIA GPU/CUDA computing environment to guarantee high throughput tensor processing. To maintain an impartial benchmarking sandbox, identical optimization hyperparameters and loss constraints were applied uniformly across all architectures:

- **Optimizer: AdamW (Adam with Weight Decay)** to stabilize transformer attention layers and prevent gradient divergence ***(adapted by I. Loshchilov [5])*.**
- **Gradient Clipping:** To improve optimization stability, global gradient clipping with a maximum L2 norm of **1.0** was applied after every backward pass. This prevented occasional gradient explosions during optimization, particularly within deeper residual networks and transformer-based architectures, while allowing stable convergence throughout the 100-epoch training schedule ***(adapted by Razvan Pascanu [7])***.
- **Loss Functions: Binary Cross-Entropy (BCE)** Loss was utilized across all independent model variants, framing the countermeasure strictly as a binary probability estimation problem between spoofed and authentic classes.
- **Spectrogram & Patch Topology:** Input audio waveforms were pre-computed into static single-channel spectrogram tensors with dimensions of **128 X 128 X 1**. For the Vision Transformer (VIT) architecture, a patch size of **16 X 16** was enforced, which yields an optimized sequence length of exactly 64 patches arranged in 8 X 8 spatial grid.
- **Base Learning Rate: $\eta = 1 \times 10^{-4}$** paired with a cosine learning rate scheduler to aggressively navigate the optimization landscape before decaying.
- **Batch Size: 32** samples per mini-batch, optimized to maximize GPU VRAM utilization and ensure highly stable gradient trajectories.
- **Regularization (Dropout and Stochastic Depth):** Dropout was applied in the relevant model components to reduce overfitting by randomly deactivating activations during training. For Transformer based architectures, DropPath (stochastic depth) was incorporated into residual branches to regularize deeper network block. The maximum DropPath rate was set to 0.1, with rates increasing across the network stages. Dropout and DropPath were disabled during inference. Regularization settings were documented according to each architecture's implementation. ***(adapted by Srivastava et al. (2014) [10]) and (adapted by Huang et al. (2016) [11])***

3.2 Evaluation Metrics: Equal Error Rate (EER) and Validation Topography

The definitive evaluation metric utilized across this empirical benchmarking framework is the **Equal Error Rate (EER)**. The EER operates as a singular operational metric derived from two conflicting error functions: the False Acceptance Rate (FAR) and the False Rejection Rate (FRR).

- **False Acceptance Rate (FAR):** The probability that the system incorrectly classifies a synthetic or manipulated acoustic sample (Spoof) as authentic (Bona Fide).

$$FAR(\theta) = \frac{\text{Number of Spoof samples accepted as Bona Fide}}{\text{Total number of Spoof samples}}$$

- **False Rejection Rate (FRR):** The probability that the system incorrectly rejects an authentic acoustic sample (Bona Fide) by classifying it as synthetic (Spoof).

$$\text{FRR}(\theta) = \frac{\text{Number of Bona Fide samples rejected as Spoof}}{\text{Total number of Bona Fide samples}}$$

The Equal Error Rate is formally defined as the specific internal threshold intersection θ_{EER} where the probability of a false acceptance exactly equals the probability of a false rejection:

$$\text{EER} = FAR(\theta_{\text{EER}}) = FRR(\theta_{\text{EER}})$$

In practice, because experimental thresholds are discrete, the EER is computationally localized by identifying the threshold θ that minimizes the absolute difference between the two rates:

$$\text{EER} \approx \min_{\theta} |FAR(\theta) - FRR(\theta)|$$

A lower EER value indicates a higher security margin and a more robust biometric anti-spoofing countermeasure, representing a well-optimized decision boundary where both types of classification errors are minimized simultaneously.

3.3 Phase 1: The Pitfalls of Premature Early Stopping

In my initial experimental phase, I applied a standard training setup utilizing a rigid early stopping monitor configured with a tight patience window (e.g stopping the run if validation loss did not improve for 15 consecutive epochs).

The empirical logs from this phase revealed a massive flaw: heavy transformer architectures were being penalized prematurely. Because transformers lack inductive bias, their early training cycles feature highly erratic validation loss plateaus. The rigid early stopping mechanism interpreted these normal optimization plateaus as a failure to converge, terminating the runs prematurely. As a result, my baseline leaderboard was dominated by the fast-converging **ResNet18 (7.46% EER)**, while the **Swin Transformer was artificially choked at an 8.35% EER**.

3.4 Phase II: Unconstrained 100-Epoch Training Strategy

Recognizing this limitation, I made the strategic decision to start training from scratch without any checkpoints. I initiated **Phase II: an unconstrained, fixed 100-epoch training block** for all architectures, entirely removing early stopping. The purpose of this experiment was to evaluate

whether allowing a consistent training duration would affect the relative performance of convolutional, Transformer-based and hybrid architectures. The resulting pre-fine tuning leaderboard was ranked using the Equal Error Rate (EER), alongside test accuracy and the area under the receiver operating characteristic curve (ROC-AUC).

The Swin Transformer achieved the lowest EER of 7.14%, with a test accuracy of 92.85% and a ROC-AUC of 0.9767. ResNet18 ranked second in terms of EER at 7.65%, achieving a test accuracy of 92.34% and a ROC-AUC of 0.9825. The CNN–Transformer hybrid obtained an EER of 9.42%, while ResNet50 recorded 9.99%. The Vision Transformer (ViT) and the two DeiT variants achieved EERs of 10.69%, 11.16%, and 11.19%, respectively.

These baseline results established the Swin Transformer as the strongest architecture in terms of EER before architecture-specific fine-tuning. Although ResNet18 achieved a slightly higher ROC-AUC, the Swin Transformer provided the lowest operating-point-independent error rate among the evaluated models. The findings also demonstrated that model depth and architectural complexity alone did not guarantee superior detection performance, highlighting the importance of empirical evaluation across different model families.

Empirical Results and Comparative Leaderboard Analysis

Following the execution of the multi-stage training framework outlined in section 3, the final architectural standings were evaluated on the independent evaluation dataset. To provide an impartial comparison, all models were evaluated using dynamic **Equal Error Rate (EER)** localization alongside standard threshold-independent area under the receiver operating characteristics curves **(ROC-AUC)** and all the pre-finetuning was done uninterrupted over 100 epochs using a learning rate of $\mathbf{1 \times 10^{-4}}$ and **32 Batch Size**. The empirical results compiled from the unified pipeline are structured in the baseline architecture leaderboard below.

Table 1. Baseline Architecture Performance Before Fine-Tuning

Model	Highest Val. Accuracy	Test. Accuracy	ROC-AUC	EER (%)
ResNet50	0.9987	0.8455	0.9873	6.47
CNN Transformer	0.9999	0.5740	0.9800	6.70
SwinTransformer_8_H	0.9898	0.8051	0.9708	7.36
SwinTransformer_4_H	0.9802	0.8294	0.9748	7.41
ResNet18	0.9969	0.8136	0.9842	7.42
DeiT-Hard	0.9636	0.8637	0.9260	13.55
DeiT-Soft	0.9628	0.8573	0.9262	13.90
VIT	0.9640	0.8634	0.9123	14.75

Note. This table summarizes baseline model performance prior to architecture-specific fine-tuning. Evaluation metrics were obtained using the common training protocol before introducing

layer-wise learning-rate decay, specialized optimizer parameter groups, or advanced fine-tuning strategies.

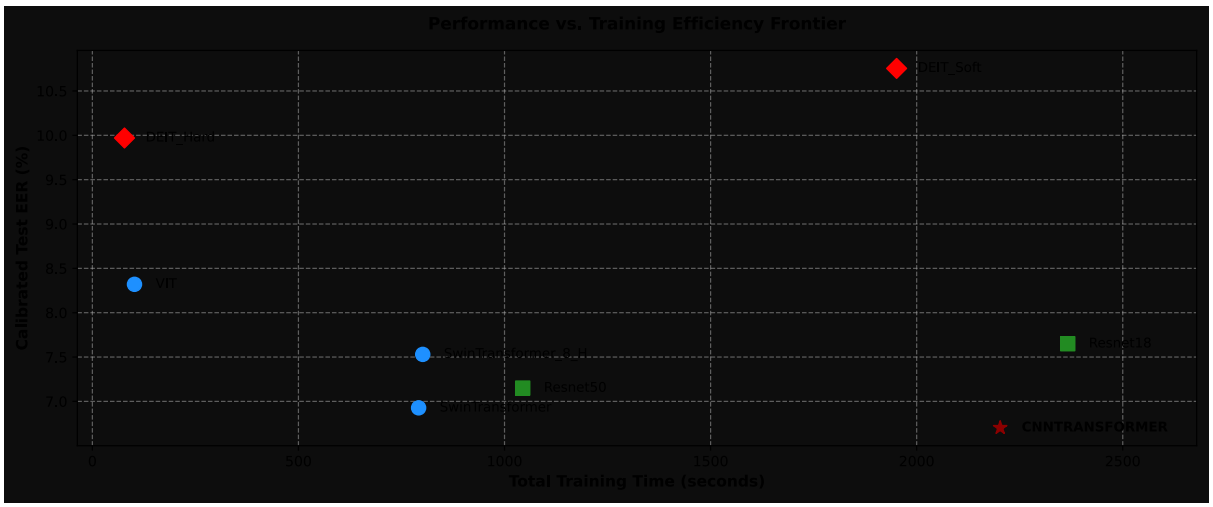


Fig 4: **EER VS Total Training Time graph shows how all models scored during pre-finetuning phase, clearly SwinTransformer was the slowest but scored the lowest EER and DEIT architecture were fast to compute but performed poorly**

4.11 Hierarchical Windowing vs Pure Convolutional

The empirical data demonstrates a competitive convergence among convolutional, hybrid and hierarchical Transformer architectures. **ResNet50** achieved the strongest overall ranking performance, recording an EER of **6.47%,** a ROC-AUC of **0.9873,** and a test accuracy of **84.55%**. its performance suggests that deeper residual convolutional feature extraction can provide an effective representation for distinguishing bona-fide and spoofed audio spectrograms under the evaluated training conditions.

The Swin Transformer variants also achieved competitive results. The eight-head Swin Transformer recorded an EER of **7.36%** and a ROC-AUC of **0.9708**, while the four-head configuration achieved an EER of **7.41%** and a ROC-AUC of **0.9748**. ResNet18 obtained an EER of **7.42%** and a ROC-AUC of **0.9842**. These results indicate that the hierarchical attention mechanism and residual convolutional architecture both provide effective discriminative representations, although ResNet50 achieved the lowest EER among the evaluated models.

The small EER differences between the Swin variants and ResNet18 suggest that architectural complexity alone does not guarantee a substantial performance improvement. Instead, the results highlight the importance of evaluating model architectures empirically using multiple performance metrics rather than assuming that deeper networks or attention-based mechanisms will consistently outperform conventional CNNs.

4.12 Hybridization and Global Projection Trade-Offs

The custom **CNN-Transformer Hybrid** achieved the second-lowest EER of **6.70%**, accompanied by a ROC-AUC of **0.9800**. This result demonstrates the potential of combining convolutional feature extraction with Transformer-based sequence modelling for audio deepfake detection. The convolutional component can capture localized spectrotemporal

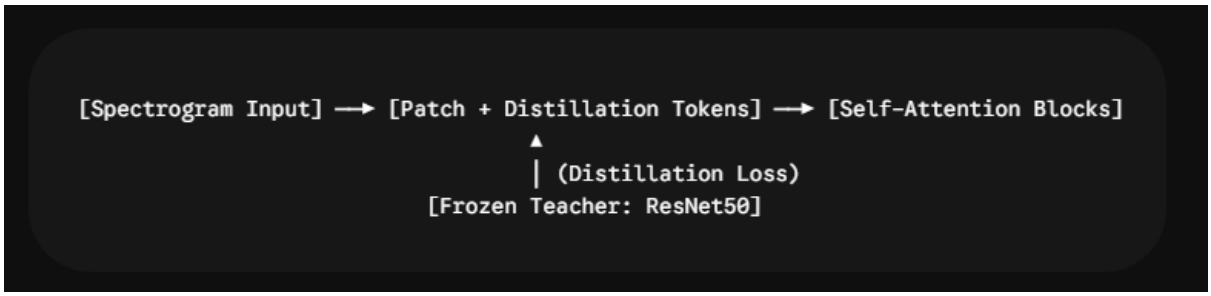
patterns, while the Transformer component models relationships across the resulting feature representations.

However, the CNN-Transformer achieved a test accuracy of only **57.40%**, considerably lower than the accuracy values of several other models despite its strong ROC-AUC and EER. This discrepancy indicates that its threshold-dependent classification performance differs markedly from its ranking performance and warrants further investigation of the decision threshold, score calibration, and evaluation implementation before drawing definitive conclusions about its operational effectiveness.

The Vision Transformer (VIT) achieved a higher EER of **14.75%** and a ROC-AUC of **0.9123**. Although its test accuracy was **83.34%** its weaker threshold-independent metrics indicate that accuracy alone does not provide a sufficient basis for comparing the discriminative capabilities of the evaluated architectures. Under the present experimental configuration, the VIT demonstrated weaker separation between bona-fide spoofed audio than the leading convolutional, hybrid, and Swin Transformer models.

4.13 Evaluation of Knowledge Distillation Constraints

The Data-Efficient Image Transformer variants (**DEIT-Hard** and **DeiT-Soft)** occupied the lower portion of the EER-based leaderboard. DeiT-Hard achieved an EER of **13.55%** and a ROC-AUC of **0.9260**, whereas DeiT-Soft recorded an EER of **13.90%** and a ROC-AUC of **0.9262**. Although their ROC-AUC values were nearly identical, DeiT-Hard achieved a slightly lower EER.



These results indicate that the two distillation configurations produced similar overall ranking performance, with a small advantage for hard distillation in terms of EER. However, the present comparison does not establish why the distillation strategies performed differently. The observed differences may reflect interactions among the teacher model, distillation objective, student architecture, and training configuration. Further controlled ablation experiments would be required to isolate the contribution of each factor.

The results also demonstrate that the inclusion of knowledge distillation does not automatically improve performance in this audio deepfake detection setting. Its effectiveness should therefore be assessed empirically rather than assumed from the original objectives of the DeiT framework.

4.14 Key Architectural Takeaways

- **Convolutional Feature Extraction Remains Competitive:** ResNet50 achieved the lowest EER of **6.47%** and the highest ROC-AUC of **0.9873**, demonstrating that conventional convolutional architecture can outperform the evaluated Transformer-based alternatives under the current experimental conditions.

- **Hybrid Architecture Shows Strong Ranking Performance:** The CNN-Transformer achieved the second lowest EER at **6.70%**. However, its substantially lower test accuracy of **57.40%** highlights the need to investigate threshold calibration and ensure consistent evaluation before interpreting its ranking performance as evidence of reliable hard classification.
- **Swin Transformer Configurations Remain Competitive:** The four-head and eight-head Swin variants achieved EERs of **7.41%** and **7.36%** respectively. Their close results suggest that increasing the number of attention heads produced only a small difference in EER under the tested configuration.
- **Accuracy and Discrimination Measure Different Aspects of Performance:** Several architectures exhibited discrepancies between test accuracy and threshold-independent metrics. For example, the DeiT variants achieved test accuracies above **85%**, despite EERs above **13%**. Consequently, ROC-AUC and EER should be examined alongside accuracy, with particular attention to threshold selection and score calibration.

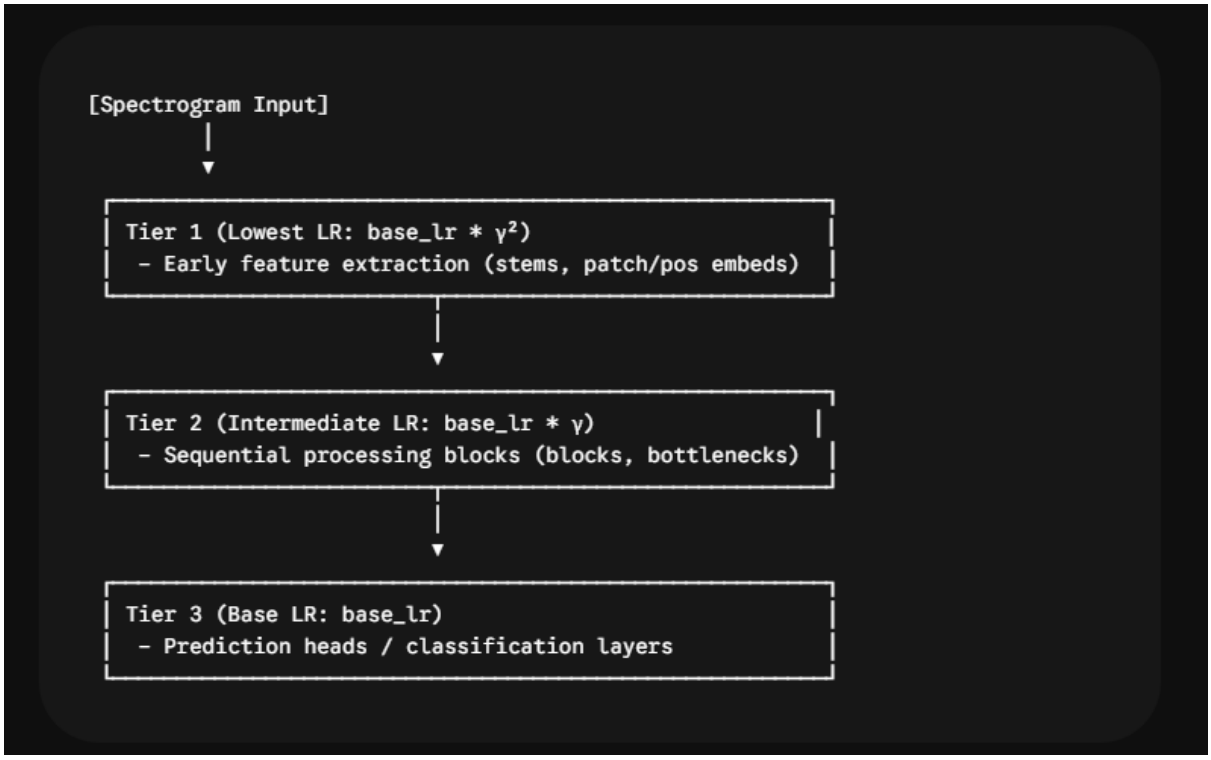
Overall, the updated baseline results identify ResNet50 as the leading architecture by EER and ROC-AUC, followed closely by the CNN-Transformer hybrid. The Swin Transformer variants and ResNet18 also demonstrated competitive discriminative performance, whereas VIT and DeiT exhibited higher error rates, These findings reinforce the importance of empirical architectural comparison and careful interpretation of both threshold-dependent and threshold-independent evaluation metrics.

5. Phase III: Fine-Tuning, Advanced Regularization, and Threshold-Compression Analysis

To push the limits of the top-performing architectures, a dedicated fine-tuning **(Phase III)** was executed. This phase targeted the optimization of **ResNet50** and the **Swin Transformer** by introducing a dynamic optimizer override framework, aggressive spectrogram data augmentation, an evaluation driven early stopping routine and learning rate warmup technique.

5.11 Multi-Tiered Optimizer Overrides and Parameter Grouping

During Phase III, the default uniform optimization configurations were replaced with specialized, multi-tiered parameter grouping framework designed to preserve generalizable features while optimizing downstream decision layers. Depending on the underlying model architecture, the pipeline dynamically applied one of three optimization strategies as discussed below:



1. **Swin Transformer LLRD:** Explicitly grouped the **patch_embed** layers into the lowest learning rate band **($\boldsymbol{LR = \eta \cdot \gamma^2}$)**, core transformer stages and normalization layers into the intermediate band **($\boldsymbol{LR = \eta \cdot \gamma}$)**, and the classification head at the base learning rate **($\boldsymbol{LR = \eta}$).**
2. **VIT and DEIT LLRD:** isolated structural positional embeddings, class tokens, and patch projection matrix into Teir 1 **($\boldsymbol{LR = \eta \cdot \gamma^2}$)**. The sequential self-attention blocks and layer norm modules were assigned to Tier 2 **($\boldsymbol{LR = \eta \cdot \gamma}$)**, while the classification projection heads were mapped to Tier **3 ($\boldsymbol{LR = \eta}$)**.
3. **Dynamic Generic Fine-Tuning Optimizer**: For all other models (ResNet18, ResNet50, and the CNN-Transformer Hybrid), a dynamic parameter parser automatically sorted parameters into a three-tiered hierarchy. Highly localized feature extractors (such as convolutional stems, early 3×3 filters, or patch projection layers) were mapped to Tier 1; deep sequential feature-processing structures (such as residual bottlenecks, attention modules, or convolutional blocks) were routed to Tier 2; and prediction heads (such as linear mapping layers or projection heads) were assigned to Tier 3.

By scaling down the base learning rate (η) sequentially from the classification head back to the earliest representation layers using a decay rate factor ($\gamma = 1.0$ for baseline fine-tuning; configured as $\gamma = 0.85$ during high-intensity sweeps), the models fine-tuned their high-level decision boundaries without destroying the foundational feature representations learned during Phase II. Additionally, to ease these parameter groups into their target distributions, a 2-epoch linear warmup phase (scaling from 10% to 100% of η) was established before transitioning into a global cosine annealing decay.

5.12 Enhanced Data Augmentation

To reduce overfitting and improve model generalization, spectrogram-domain data augmentation was incorporated during fine-tuning. Dynamic frequency masking was applied to randomly mask frequency bands, encouraging the models to learn robust spectral representations. In addition, Gaussian noise was added to the input spectrograms with a low

noise variance, simulating real-world recording noise and preventing the models from memorizing dataset-specific acoustic patterns. Additionally, lossy-compression augmentation was introduced to approximate distortions associated with compressed audio, including simulated high-frequency attenuation and quantization of spectrogram values. These augmentations improved robustness by exposing both convolutional and transformer architectures to a wider range of acoustic variations during training.

5.13 EER-Driven Early Stopping

Rather than monitoring traditional Validation Loss or Validation Accuracy- which are highly susceptible to threshold misalignment-the early stopping hook inside **Trainer.py** was reengineered to monitor the **Equal Error Rate (EER)** directly on the validation partition:

```
# Track and stop based on actual EER improvements
if current_val_eer < self.best_val_eer or epoch == self.start_epoch:
    self.best_val_eer = current_val_eer
    patience_counter = 0  # Reset early stopping clock
    self.save_checkpoint(epoch, val_acc)
else:
    patience_counter += 1
```

By calculating the dynamic ***EER*** at the end of each epoch and establishing a patience window of 10 epochs, the pipeline successfully terminated training at the absolute peak of the model's discriminative capability, preserving its generalization boundary.

5.14 Learning Rate Scheduler

During the fine-tuning phase, a sequential learning rate scheduling strategy combining linear warmup and cosine annealing was employed. The warmup phase lasted two epochs, during which the learning rate was linearly increased from 10% to 100% of its initial value to facilitate a gradual transition into fine-tuning and reduce abrupt parameter updates at the beginning of optimization. Following warmup, a Cosine Annealing Learning Rate (CALR) scheduler was applied for the remaining training epoch, progressively reducing the learning rate toward a minimum value of (1 \times 10^{-7). This strategy was designed to support stable optimization during the initial fine-tuning phase while enabling gradual convergence during subsequent training, following the cosine annealing approach ***(adapted by I. Loshchilov [8])***.

5.2 Fine-Tuned Empirical Results

Following the application of the fine-tuning strategy, the comparative leaderboard was updated to evaluate the performance of the eight architectures on the **ASVspoof2019 Logical Access (LA)** evaluation set. The fine-tuning procedure is incorporated architecture-specific optimizer configurations, enhanced spectrogram-domain data augmentation, a two-epoch linear learning-rate warmup followed by cosine annealing, and EER-driven early stopping. Model performance was assessed using evaluation accuracy, the area under the receiver operating characteristic curve (ROC-AUC), and Equal Error Rate (EER). The results are summarized in Table 2, with the architectures ranked according to their reported EER, where lower values indicate better score-based discrimination between bona-fide and spoofed speech.

Table 2. Best Observed Coherent Evaluation Performance Following Architecture-Specific Fine-Tuning

Model	Highest Val. Accuracy	Test. Accuracy	ROC-AUC	EER (%)
SwinTransformer_8_H	0.9687	0.9195	0.9844	5.86
SwinTransformer_4_H	0.9802	0.9184	0.9826	5.97
ResNet18	0.9999	0.8233	0.9864	6.01
ResNet50	0.9987	0.8455	0.9873	6.47
CNN Transformer	0.9999	0.5740	0.9800	6.70
DeiT-Soft	0.9357	0.9050	0.9726	8.21
DeiT-Hard	0.9967	0.8007	0.9718	8.76
VIT	0.9945	0.8168	0.9585	10.39

Note. Lower EER is better. The table reports best-observed coherent evaluation fine tuned runs

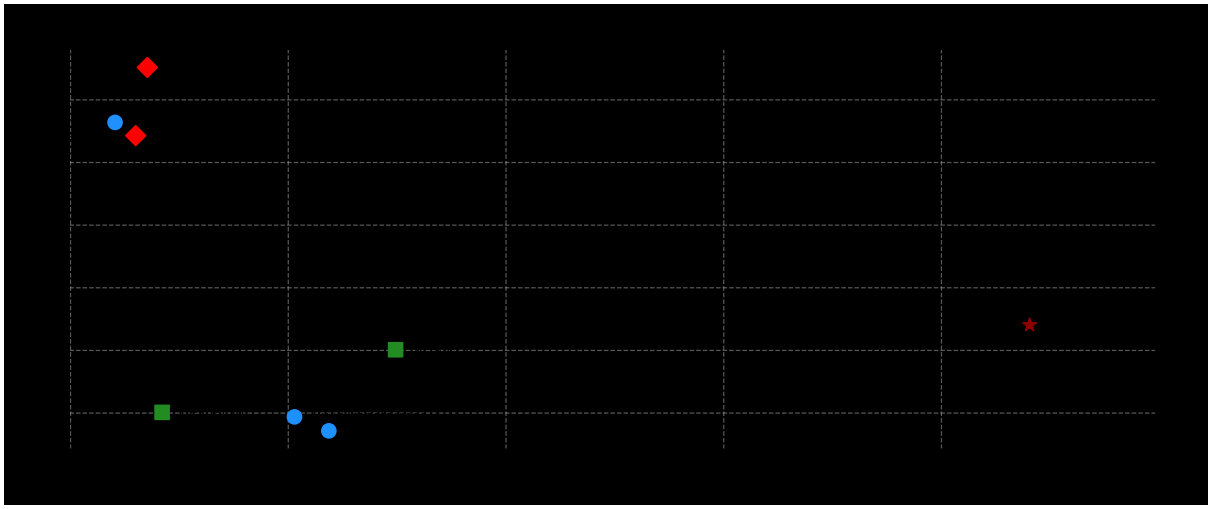


Fig 5: **EER vs Training Time for Fine Tuned models, as we can see almost all the models EER value improved with fine tuning methods and training time decreased too except Swin Transformer as the prediction collapsed**

5.2.1 Performance of Swin Transformer Architectures

The Swin Transformer with eight attention heads achieved the strongest overall result in the updated leaderboard, recording an EER of **5.86%**, an evaluation accuracy of **91.95%**, and a ROC-AUC of **0.9844**. The four-head configuration achieved closely comparable performance, with an EER of **5.97%**, an evaluation accuracy of **91.84%**, and a ROC-AUC of **0.9826**. These results place both Swin configurations among the strongest evaluated architectures following fine-tuning.

The difference in EER between the eight-head and four-head configurations was only 0.11 percentage points. This relatively small difference suggests that increasing the number of attention heads provided a modest advantage under the evaluated training configuration rather than a substantial improvement in performance, Both configurations maintained high ROC-AUC values and evaluation accuracies above 91% indicating strong discrimination on the source-domain evaluation set.

The results are consistent with the potential benefits of hierarchical attention for spectrogram-based classification. The Swin Transformer partitions feature maps into local windows and uses shifted window operations to facilitate information exchange between neighboring regions. Such a design can model localized time-frequency patterns while progressively integrating broader contextual information. Nevertheless, the leaderboard alone cannot establish this mechanism directly caused the performance improvement. A more definitive causal comparison would require controlled experiments that isolate architectural differences while holding the remaining training settings constant.

5.2.2 Comparison of Residual Convolutional Architectures

ResNet18 and ResNet50 also demonstrated competitive performance following fine-tuning. ResNet18 achieved an EER of **6.01%,** an evaluation accuracy of **82.23%**, and a ROC-AUC of **0.9864**. ResNet50 recorded an EER of **6.47%**, an evaluation accuracy of **84.55%**, and a ROC-AUC of **0.9873,** which doesn't suggest an improvement from pre fine tuning phase for ResNet50. Although neither model surpassed the eight-head Swin Transformer in EER, both

residual networks produced strong ROC-AUC values, indicating a high degree of separation between the class score distributions on the evaluation set.

ResNet50 achieved the highest ROC-AUC among all eight models, at **0.9873**, while ResNet18 recorded a lower EER. This difference illustrates that ROC-AUC and EER capture related but distinct aspects of score-based discrimination. ROC-AUC measures ranking performance across decision thresholds, whereas EER identifies the operating point at which the false acceptance and false rejection rates are approximately equal. Consequently, the architecture with the highest ROC-AUC does not necessarily achieve the lowest EER.

The comparison also shows that increasing network depth did not uniformly improve every metric. Despite its greater depth and representational capacity, ResNet50 recorded an EER **0.46** percentage points higher than ResNet18, although it achieved a **0.0009** higher ROC-AUC and a **2.32**-percentage-point higher evaluation accuracy. These differences suggest that the relative benefits of deeper convolutional representations depend on the performance criterion being examined. The results therefore support evaluating residual architectures using several complementary metrics rather than relying on model depth or a single performance measure.

5.2.3 Performance of the CNN-Transformer Hybrid

The custom CNN–Transformer architecture achieved an EER of **6.70%** and a ROC-AUC of **0.9800**, placing it fifth in the leaderboard according to EER. These values indicate strong score-based discrimination relative to several of the other evaluated models. The hybrid architecture combines convolutional feature extraction with Transformer-based contextual modelling, allowing local spectrogram patterns to be processed before their relationships are modelled through attention mechanisms. However, there was no improvement in metrics from pre finetuning phase.

However, its evaluation accuracy was only 57.40%, substantially below that of the Swin Transformer configurations and both residual convolutional models. This discrepancy is important because it indicates that the classification performance obtained at the evaluation threshold does not correspond closely to the model's ranking-based metrics. A low EER and a high ROC-AUC indicate that useful score ordering may be present, but they do not guarantee high accuracy at a particular fixed threshold.

The difference warrants additional verification of the CNN–Transformer's threshold calibration, output convention, and evaluation implementation. In particular, the classification threshold must be applied consistently to the intended spoof probability, and the accuracy, ROC-AUC, and EER must be computed from the same samples and checkpoint. Until this discrepancy is resolved, the CNN–Transformer should be described as having strong reported ranking performance but inconsistent threshold-dependent classification performance, rather than as an unequivocally superior practical detector.

5.2.4 Evaluation of VIT and DEIT Architectures

The DeiT variants demonstrated lower ranking performance than the leading Swin and residual architectures. DeiT-Soft achieved an EER of **8.21%**, an evaluation accuracy of **90.50%**, and a ROC-AUC of **0.9726**. DeiT-Hard recorded an EER of **8.76%**, an evaluation accuracy of **80.07%**, and a ROC-AUC of **0.9718**. The soft-distillation configuration therefore achieved a 0.55-percentage-point lower EER and a 10.43-percentage-point higher evaluation accuracy than the hard-distillation configuration, while their ROC-AUC values remained close.

These results suggest that the soft-distillation configuration performed better under the reported evaluation protocol. However, they do not establish that soft distillation is inherently more effective for audio deepfake detection. The relative performance may depend on the teacher model, the distillation loss, the student architecture, and the optimization configuration. Further controlled experiments would be required to attribute the difference to the distillation strategy itself.

The Vision Transformer (ViT) recorded the weakest overall ranking performance among the evaluated models, with an EER of **10.39%**, a ROC-AUC of **0.9585**, and an evaluation accuracy of **81.68%**. Its higher EER indicates weaker separation between bona-fide and spoof scores compared with the Swin, residual, and CNN–Transformer architectures in the current experiment. One possible explanation is that hierarchical local processing provides a useful inductive structure for spectrogram representations when training data and input resolution are limited. However, this remains an interpretation rather than a causal conclusion established by the leaderboard alone.

The ViT and DeiT results indicate that Transformer-based architectures did not perform uniformly across the tested configurations. While the Swin variants achieved the two lowest EER values, the plain ViT and distilled DeiT variants recorded higher error rates. This contrast reinforces the importance of architectural design and training configuration in determining performance, even among models derived from related Transformer-based principles.

5.2.5 Effect of Fine-Tuning on Baseline Performance

Comparison with the previously reported pre-fine-tuning results indicates that several architectures achieved lower EER after fine-tuning. ResNet18 improved from a baseline EER of 7.65% to 6.01%, representing a reduction of 1.64 percentage points. ResNet50 and CNN Transformer Hybrid metrics didn't improve during fine tuning phase.

The DeiT variants showed the most improvement during fine tuning phase with the models achieving lower EERs. DeiT-Hard decreased from 13.55% to 8.76%, while DeiT-Soft decreased from 13.90% to 8.21%. In comparison, the ViT showed a reduction from 14.75% to 10.39%. These results indicate that the magnitude of the observed improvement varied across architectures, with the largest reductions among these directly comparable baseline entries occurring for ResNet50 and CNN–Transformer.

The newer leaderboard additionally distinguishes the eight-head and four-head Swin configurations, recording EERs of 5.86% and 5.97%, respectively. Because these configurations are listed separately from the earlier general Swin Transformer baseline, their results should be interpreted as architecture-specific experimental outcomes rather than treated as a strictly controlled before-and-after comparison unless the corresponding baseline configurations and training protocols are matched.

Although the observed EER reductions are consistent with improved source-domain score discrimination, they should not be interpreted as conclusive evidence that every fine-tuning component independently improved performance. The experiments changed several optimization and augmentation settings, so the contribution of each modification cannot be isolated without dedicated ablation studies. Furthermore, threshold-dependent evaluation accuracy did not improve uniformly across all models, emphasizing the need to interpret the results using both threshold-dependent and threshold-independent metrics.

5.2.6 Summary of Architectural Findings

Overall, the updated ASVspoof2019 LA leaderboard identifies Swin Transformer with eight attention heads as the strongest model by EER, followed closely by Swin Transformer with four attention heads and ResNet18. ResNet50 achieved the highest ROC-AUC, demonstrating that different architectures can lead on different performance measures. The DeiT variants achieved lower EERs than in their previously reported baseline results, although the CNN–Transformer's unusually low evaluation accuracy requires further investigation.

The results demonstrate that both convolutional and Transformer-based approaches can provide effective score-based discrimination for audio deepfake detection, but that the magnitude of their benefits depends on architecture and evaluation metric. They also show that high validation accuracy is not sufficient to establish strong classification performance, particularly when threshold calibration and score distributions differ. These source-domain results provide the basis for the subsequent cross-dataset experiments, which examine whether the observed architecture rankings and discriminative performance are maintained when the models are evaluated on datasets outside ASVspoof2019 LA.

5.3 Cross-Dataset Evaluation

To investigate whether the performance observed on ASVspoof2019 LA translates to previously unseen evaluation datasets, the trained architectures were evaluated on ASVspoof2021 DF, In-the-Wild and the LJSpeech-MelGAN subset of WaveFake. These datasets introduced different speech-generation methods, source recordings, and audio-processing conditions, providing complementary perspectives on the generalization capabilities of the evaluated models. The models were evaluated using their existing trained checkpoints without updating their weights on the external test data and zero shot experiment was used. The same log-mel spectrogram preprocessing pipeline was maintained to ensure consistency with the source-domain experiments. ROC-AUC and Equal Error Rate (EER) were used to assess score-based discrimination, while classification accuracy was reported using the decision threshold established during ASVspoof2019 validation. This evaluation protocol enables a comparison between in-domain performance and external-dataset generalization, including the extent to which source-domain decision thresholds remain effective under distribution shift. ***(adapted by Liu*** [12]***)***, ***(adapted by Joel Frank*** [13]***)***, ***(adapted by Nicolas Muller*** [14])

5.31 ASVspoof2021 DF Dataset Evaluation

Table 3. ASVspoof2021 DF Evaluation Performance Following Architecture-Specific Fine-Tuning

Model	Highest Val. Accuracy	Test. Accuracy	ROC-AUC	EER (%)
CNN Transformer	1.0000	0.6840	0.8792	20.70
SwinTransformer_8_H	0.9879	0.3916	0.8883	20.75
SwinTransformer_4_H	0.9896	0.4409	0.8876	20.83
VIT	0.9919	0.6673	0.8338	24.09
DeiT-Hard	0.9975	0.7502	0.8369	24.30
ResNet50	0.9978	0.7819	0.8497	24.70
DeiT-Soft	0.9967	0.7125	0.8327	24.92
ResNet18	0.9998	0.7816	0.7609	32.93

Evaluation on ASVspoof2021 DF revealed a substantial reduction in performance compared with the ASVspoof2019 LA results. The CNN–Transformer hybrid achieved the lowest EER of 20.70%, with a ROC-AUC of 0.8792 and an evaluation accuracy of 68.40%, validation accuracy of 100% which doesn't indicate that model was overfitting or has memorized data it had converged during training with ASVspoof2019 LA dataset. Swin Transformer with eight attention heads followed closely, recording an EER of 20.75% and the highest ROC-AUC of 0.8883, although its evaluation accuracy was considerably lower at 39.16%. The four-head Swin configuration achieved a comparable EER of 20.83% and a ROC-AUC of 0.8876. In contrast, ResNet18 exhibited the weakest generalization among these leading comparisons, with an EER of 32.93% and a ROC-AUC of 0.7609.

The degradation in performance cannot be explained simply by the larger number of evaluation samples in ASVspoof2021 DF. Although a larger evaluation set can provide a more comprehensive assessment of model performance, these additional samples do not improve a model that has already been trained exclusively on ASVspoof2019 LA. Instead, the more important factor is the difference between the training and evaluation distributions. ASVspoof2021 DF was designed to evaluate robustness under less controlled conditions, incorporating compressed audio, multiple codec conditions, and additional speech sources intended to test domain generalization. Such processing can alter or attenuate the spectral artefacts that a detector learns to associate with synthetic speech. Consequently, features that are highly discriminative on ASVspoof2019 LA may become less reliable when evaluated on the more varied audio conditions of ASVspoof2021 DF. ***(adapted by Liu*** [12]***)***

The results also reveal differences between score-based discrimination and threshold-dependent classification. Although the CNN–Transformer achieved the lowest EER, the eight-head Swin Transformer obtained the highest ROC-AUC, suggesting that the architectures differ slightly in their ranking performance. The substantially lower evaluation accuracy reported for

the eight-head Swin configuration further indicates that a decision threshold established on the source dataset may not transfer effectively to the external evaluation set. These results demonstrate that cross-dataset generalization depends not only on the architecture's ability to rank bona-fide and spoofed samples, but also on whether its decision threshold remains appropriate under distribution shift.

Overall, the findings show that strong in-domain performance on ASVspoof2019 LA does not guarantee equivalent performance on ASVspoof2021 DF. The results emphasize the importance of evaluating audio deepfake detectors under varying compression conditions and source distributions, rather than relying solely on the number of samples or the high validation accuracy achieved during training.

5.32 In-the-Wild Dataset Evaluation

Table 4. In-the-Wild Dataset Evaluation Performance Following Architecture-Specific Fine-Tuning

Model	Highest Val. Accuracy	Test. Accuracy	ROC-AUC	EER (%)
CNN Transformer	1.0000	0.3570	0.5175	47.36
ResNet18	0.9998	0.3707	0.5438	47.36
ResNet50	0.9996	0.3697	0.4544	52.02
SwinTransformer_8_H	0.9981	0.3801	0.4459	54.70
SwinTransformer_4_H	0.9692	0.3907	0.4367	54.85
DeiT-Hard	0.9975	0.3734	0.4183	55.48
DeiT-Soft	0.9967	0.3569	0.3997	56.69
VIT	0.9921	0.3486	0.3159	63.86

The In-the-Wild evaluation produced the largest performance degradation among the evaluated datasets. ResNet18 and the CNN–Transformer hybrid achieved the lowest reported EER, at 47.36%, but their ROC-AUC scores were only 0.5438 and 0.5175, respectively. ResNet18 recorded an evaluation accuracy of 37.07%, while the CNN–Transformer achieved 35.70%. ResNet50 performed less effectively, with an EER of 52.02% and a ROC-AUC of 0.4544. The Transformer-based models also exhibited substantial difficulties: Swin Transformer 8H achieved an EER of 54.70%, while Swin Transformer 4H recorded 54.85%. DeiT-Hard and DeiT-Soft obtained EERs of 55.48% and 56.69%, respectively, whereas ViT showed the weakest discrimination, with an EER of 63.86% and a ROC-AUC of 0.3159.

These results indicate that none of the evaluated architectures achieved reliable separation between bona-fide and spoofed speech in the reported In-the-Wild experiment. Although ResNet18 and the CNN–Transformer obtained the lowest EER values, their results remained close to chance-level discrimination. Furthermore, the ROC-AUC values below 0.5 for several architectures indicate that their score rankings were, on average, opposite to the expected bona-fide-versus-spoof ordering. This behaviour warrants verification of score polarity, label

conventions, and the numerical stability of the evaluation metrics before definitive conclusions are drawn about the relative performance of these models.

Influence of Real-World Recording Conditions

A major difference between ASVspoof2019 LA and In-the-Wild is the origin of the audio recordings. ASVspoof2019 LA uses genuine speech from the VCTK corpus, which was recorded in a controlled studio environment, alongside spoofed samples generated using a defined set of synthesis and voice-conversion systems. In contrast, In-the-Wild contains genuine and deepfake speech associated with 58 celebrities and politicians, collected from publicly available sources such as social networks and video-sharing platforms ***(adapted by Nicolas Muller*** [14]).

Consequently, In-the-Wild presents a different acoustic environment from the source dataset. Online-video audio can be affected by compression, transcoding, background noise, room reverberation, music, microphone characteristics, and additional audio or video editing. Such conditions may obscure or alter subtle spectrotemporal patterns learned by a detector during training. A model that relies heavily on synthesis artefacts observed in ASVspoof2019 may therefore become less effective when those artefacts are attenuated, modified, or masked by subsequent processing ***(adapted by Nicolas Muller*** [14]).

The distinction should not, however, be attributed solely to lower audio quality. The original In-the-Wild dataset was designed to represent realistic recordings, and its authors took care to collect corresponding genuine material with broadly comparable speaker characteristics, speaking styles, emotions, and background conditions. The observed performance reduction is therefore better interpreted as a combination of acoustic variability and differences between the training and evaluation domains, rather than simply as a consequence of noisier recordings.

Effect of Audio Representation and Preprocessing

The models in this study received single-channel log-mel spectrograms with a fixed input size of (128 \times 128). Audio was resampled to 16 kHz and converted to mono during preprocessing. This maintains a consistent tensor format across datasets but does not eliminate differences in the underlying acoustic information. In particular, reducing the sample rate and converting recordings to a common channel configuration cannot reverse the effects of compression, recover removed frequency detail, or remove background interference.

The fixed spectrogram dimensions may introduce an additional consideration. Recordings that exceed the retained time window are truncated, whereas shorter recordings are padded to the target duration. These operations can alter the amount and distribution of useful speech information presented to the network. Because the In-the-Wild clips have variable durations and originate from heterogeneous sources, it is important to verify that the fixed-length representation does not introduce systematic differences between the two classes. The padding convention should also be inspected carefully: in the implemented preprocessing pipeline, padding is applied after conversion to decibels, so the default padding value of zero does not represent silence relative to the spectrogram's maximum reference level.

These factors are plausible contributors to the observed generalization gap, but the present results do not isolate their individual effects. Further experiments controlling for compression,

duration, and preprocessing would be necessary to determine how much each factor contributes to the reduction in detection performance.

Architecture Performance and Decision-Threshold Transfer

The architecture rankings on In-the-Wild differed considerably from those on ASVspoof2019 LA and ASVspoof2021 DF. ResNet18 and the CNN–Transformer produced the lowest reported EERs, whereas the Swin Transformer variants, DeiT models, and ViT performed substantially worse. This indicates that the architectural advantages observed in the source-domain evaluation did not transfer uniformly to the real-world dataset.

Evaluation accuracy also requires careful interpretation. The In-the-Wild dataset contains more bona-fide than spoofed recordings, and the reported accuracies of approximately 35–39% are consistent with predictions biased heavily toward the spoof class. This indicates that the decision thresholds established during ASVspoof2019 validation did not transfer effectively to the external score distributions. However, threshold mismatch alone cannot explain the high EER values or the low ROC-AUC scores. Those metrics indicate that limited class separation, and potentially reversed score ordering in some models, must also be considered.

The results align with the broader findings of Müller et al. ***(adapted by Nicolas Muller*** [14]), who reported substantial performance degradation when established audio deepfake detectors were evaluated on their In-the-Wild dataset. The present experiment similarly demonstrates that performance on controlled benchmark data should not be assumed to represent effectiveness on audio obtained from unconstrained public sources.

Overall, In-the-Wild proved to be the most challenging evaluation environment for the architectures examined in this study. The results highlight the limitations of relying solely on in-domain validation accuracy and demonstrate the importance of external testing across recording environments and synthesis methods. They also motivate further investigation into alternative audio representations, more robust score calibration, and training strategies that improve generalization to real-world speech.

5.33 WaveFake Dataset Evaluation

Table 5. WaveFake Dataset Evaluation Performance Following Architecture-Specific Fine-Tuning

Model	Highest validation accuracy	Test accuracy	ROC-AUC	EER (%)
Swin Transformer 8H	0.9951	0.5001	0.7365	31.63
Swin Transformer 4H	0.9885	0.5001	0.7008	34.25
CNN–Transformer	0.9998	0.5000	0.6735	37.54
DeiT-Soft	0.9967	0.5059	0.5981	43.37
ViT	0.9861	0.5220	0.5827	44.23
DeiT-Hard	0.9949	0.5048	0.5729	45.01
ResNet18	0.9998	0.5002	0.4808	51.51

Evaluation on the WaveFake LJSpeech–MelGAN subset revealed a substantial reduction in performance compared with the ASVspoof2019 LA and ASVspoof2021 DF results. Swin Transformer with eight attention heads achieved the strongest overall performance, recording an EER of 31.63% and a ROC-AUC of 0.7365, although its evaluation accuracy was only 50.01%. The four-head Swin Transformer configuration followed with an EER of 34.25% and a ROC-AUC of 0.7008. The CNN–Transformer hybrid obtained an EER of 37.54% and a ROC-AUC of 0.6735. In contrast, DeiT-Soft, ViT, and DeiT-Hard recorded EERs of 43.37%, 44.23%, and 45.01%, respectively. ResNet18 and ResNet50 exhibited the weakest performance, with EERs of 51.51% and 53.75% and ROC-AUC scores of 0.4808 and 0.4486, respectively.

Despite achieving source-domain validation accuracies between 98.61% and 99.98%, all models recorded evaluation accuracies close to 50%. This discrepancy indicates that the high validation accuracy obtained during training did not translate into reliable classification decisions on the WaveFake evaluation set. The results suggest that the evaluated models struggled to transfer their learned representations to MelGAN-generated speech, highlighting a substantial difference between their performance on the training domain and their ability to generalize to an unseen synthesis system.

Influence of Generator-Specific Characteristics

An important distinction between WaveFake and the earlier evaluation datasets is the synthesis system used to generate the spoofed speech. ASVspoof2019 LA contains spoofed recordings generated using a defined collection of speech synthesis and voice-conversion systems, whereas the current WaveFake evaluation focuses specifically on speech generated using MelGAN. Consequently, the evaluation examines whether features learned from ASVspoof2019 LA remain useful when the detector encounters a different vocoder. ***(adapted by Joel Frank*** [13]***)***

The WaveFake subset used in this study contains 13,100 bona-fide LJSpeech recordings and 13,100 corresponding MelGAN-generated recordings, giving a balanced evaluation set of

26,200 samples. Because the genuine and generated recordings correspond to the same underlying utterances, differences in linguistic content and speaker identity are reduced as potential confounding factors. This structure makes the dataset particularly useful for examining whether a detector can identify differences associated with synthetic speech rather than relying primarily on variations in spoken content. ***(adapted by Joel Frank*** [13]***)***

Nevertheless, a matched utterance structure does not eliminate all sources of distribution shift. MelGAN can introduce characteristic spectral patterns that differ from those produced by synthesis systems represented in ASVspoof2019 LA. If a model has learned to recognize generator-specific artefacts during training, those features may provide limited information when the model encounters a previously unseen vocoder. The observed degradation is consistent with this possibility, although the current experiments do not independently establish that generator-specific artefacts are the primary cause. ***(adapted by*** *Konstantin Gasenzer* [15])

The architecture rankings provide further evidence of differences in cross-dataset transferability. Swin Transformer 8H retained the strongest discriminative performance, followed by Swin Transformer 4H and CNN–Transformer. The residual networks, which performed relatively well on ASVspoof2019 LA, performed particularly poorly on WaveFake. These differences suggest that the effectiveness of an architecture depends not only on its ability to distinguish bona-fide and spoofed speech within the source dataset, but also on whether the features it learns remain informative for unseen synthesis methods.

Evaluation Accuracy, ROC-AUC, and Decision-Threshold Transfer

A notable feature of the WaveFake leaderboard is the discrepancy between evaluation accuracy and the score-based metrics. Although Swin Transformer 8H achieved a ROC-AUC of 0.7365, its evaluation accuracy was only 50.01%. A similar pattern appeared across the remaining architectures, whose evaluation accuracies ranged from 50.00% to 52.20%.

Since the WaveFake evaluation set contains equal numbers of bona-fide and spoofed samples, a classifier that predicts the same class for every recording would achieve approximately 50% accuracy. The reported accuracies are therefore consistent with the possibility that the models' decisions were heavily concentrated on one class.

Nevertheless, threshold mismatch alone cannot explain the full performance reduction. Swin Transformer 8H recorded an EER of 31.63%, indicating substantial remaining classification error even when the threshold is adjusted to balance the false acceptance and false rejection rates on the evaluation set. The increase in EER, alongside the reduction in ROC-AUC, therefore indicates that the models experienced both a deterioration in discrimination and a potential problem transferring their source-domain decision thresholds.

ResNet18 and ResNet50 produced ROC-AUC values below 0.5. Under the current score convention, these results indicate that their scores generally ranked the classes in the opposite direction from the expected ordering. This behaviour is consistent with particularly weak discrimination under the current evaluation setup, rather than reliable classification performance. The results should be interpreted using the same class-label and score-polarity convention across all datasets.

Effect of Audio Representation and Preprocessing

All models received single-channel log-mel spectrograms with a fixed input size of \(128 \times 128\). The audio was resampled to 16 kHz and converted to mono before being transformed into the model input representation. This consistent preprocessing pipeline ensures that each architecture receives a compatible tensor format across datasets, but it does not eliminate differences introduced by the original recording or synthesis processes.

In particular, MelGAN-generated speech may contain spectral characteristics that differ from those of the spoofed samples represented in ASVspoof2019 LA. A fixed log-mel representation may preserve some of these differences while obscuring others. Consequently, the learned features may be informative for the synthesis methods encountered during training but less effective for the particular artefacts present in the MelGAN recordings.

The fixed-length preprocessing also warrants investigation. Recordings longer than the retained input window are truncated, whereas shorter recordings are padded to the target duration. These operations can affect the distribution of speech and non-speech information presented to the network. In the implemented pipeline, padding is applied after conversion to decibels, meaning that the default zero-padding value corresponds to 0 dB rather than a low-energy representation of silence. This may introduce conspicuous regions into the spectrograms. If the proportion of padded frames differs systematically between genuine and spoofed recordings, a detector could potentially exploit these differences as shortcuts.

These preprocessing considerations are plausible contributors to the observed generalization gap, but the current leaderboard does not establish their individual effects. Further experiments comparing duration distributions, padding fractions, and predictions under alternative preprocessing strategies would be needed to determine whether such artefacts influence the WaveFake results.

Architecture Performance and Potential Shortcut Learning

The combination of high source-domain validation accuracy and substantially weaker external performance raises the possibility that some models learned features specific to the ASVspoof2019 LA dataset rather than broadly generalizable characteristics of synthetic speech. This behaviour is consistent with shortcut learning, in which a model relies on features that correlate with the target labels in its training data but are unreliable under distribution shift. ***(adapted by Nicolas Muller*** [14]), ***(adapted by*** *Konstantin Gasenzer* [15])

For example, a detector may learn spectral artefacts associated with the synthesis systems represented in ASVspoof2019 LA, as well as differences in duration, silence, or preprocessing. When evaluated on MelGAN-generated speech, these cues may be absent, altered, or less informative. The resulting decline in performance would be consistent with a model relying on generator-specific or dataset-specific characteristics. ***(adapted by*** *Konstantin Gasenzer* [15])

The WaveFake results are compatible with this hypothesis, particularly given the differences in performance across architectures. Swin Transformer 8H retained a ROC-AUC above 0.7, suggesting that it preserved some ability to rank the two classes, whereas ResNet18 and ResNet50 performed poorly under the current score convention. However, the leaderboard alone cannot demonstrate that shortcut learning occurred or identify which features the models relied on. Generator-specific domain shift, preprocessing effects, and other differences in the data distributions remain plausible explanations.

To establish whether shortcut learning contributes to the observed performance degradation, additional analysis would be required. This could include comparing spectrograms and frequency-domain characteristics across datasets, evaluating the correlation between predictions and preprocessing artefacts, and testing whether performance changes when duration, padding, and other potentially confounding features are controlled.

5.34 WaveFake Comparison with ASVspoof2019 LA, ASVspoof2021 DF and In-the-Wild

Evaluation Dataset	Lowest EER	Model Achieving lowest EER	Highest ROC-AUC	Model achieving highest ROC-AUC
ASVspoof2019 LA	6.47%	ResNet50	0.9873	ResNet50
ASVspoof2021 DF	20.70%	CNN-Transformer	0.8883	Swin Transformer_8_H
WaveFake	31.63%	Swin Transformer_8_H	0.7365	Swin Transformer_8_H
In-the-Wild	47.36%	CNN-Transformer and ResNet18	0.5438	ResNet18

The progression from an EER of 6.47% on ASVspoof2019 LA to 20.70% on ASVspoof2021 DF and 31.63% on WaveFake demonstrates a substantial reduction in detection performance as the models are evaluated on different datasets. While ASVspoof2021 DF introduces more challenging audio conditions, including compression and codec-related variation, WaveFake tests the models against a different synthesis system. The latter therefore provides complementary evidence that the learned representations do not transfer consistently across spoofing domains.

The decline in ROC-AUC further supports this observation. The strongest ROC-AUC decreased from 0.9873 on ASVspoof2019 LA to 0.8883 on ASVspoof2021 DF and 0.7365 on WaveFake. The results indicate that the score distributions became less effective at ranking bona-fide samples above spoofed samples, assuming the same positive-class and score-direction convention across evaluations. Model also had to deal with smoking gun problem.

In-the-Wild presented an even greater challenge, with the best ROC-AUC of only 0.5438 and a lowest reported EER of 47.36%. Unlike the controlled, matched-utterance comparison in WaveFake, In-the-Wild contains heterogeneous recordings collected from public sources, introducing additional variation in speakers, recording environments, processing, and synthesis methods. The difference between these results suggests that the WaveFake evaluation is challenging primarily in terms of transfer to a different synthesis domain, whereas In-the-Wild combines synthesis-related differences with greater real-world acoustic variability.

The architecture rankings also changed across datasets. ResNet50 achieved the lowest EER on ASVspoof2019 LA, CNN–Transformer achieved the lowest EER on ASVspoof2021 DF, and Swin Transformer 8H performed best on WaveFake. No architecture was consistently superior across all evaluations. This finding indicates that in-domain validation accuracy and

benchmark performance alone are insufficient to determine which architecture will generalize most reliably to unfamiliar audio sources.

Overall Findings

Overall, the WaveFake evaluation demonstrated that the strong source-domain performance achieved on ASVspoof2019 LA did not translate directly into reliable detection of MelGAN-generated speech. Swin Transformer 8H achieved the strongest results, with an EER of 31.63% and a ROC-AUC of 0.7365, followed by Swin Transformer 4H and CNN–Transformer. However, the evaluation accuracies of approximately 50% and the comparatively high EER values indicate that all architectures struggled to make reliable classification decisions at the applied operating thresholds.

Compared with ASVspoof2021 DF, WaveFake produced lower ROC-AUC values and higher EERs, suggesting that transfer to an unseen synthesis system introduced additional challenges. In-the-Wild proved even more difficult overall, likely reflecting a combination of synthesis-domain differences and greater variability in real-world audio sources.

These findings emphasize the importance of evaluating audio deepfake detectors across multiple datasets and synthesis systems rather than relying exclusively on high in-domain validation accuracy. They also motivate further investigation into generator-independent representations, consistent preprocessing, score calibration, and possible shortcut learning. The current results establish a substantial cross-dataset generalization gap, while the precise causes of that gap remain to be determined.

5.35 Lessons Learned from Fine-Tuning and Cross-Dataset Evaluation

Limitations of Source-Domain Validation Performance

One of the principal findings from the experiments is that high validation accuracy does not necessarily indicate that an audio deepfake detector will generalize effectively to unseen datasets. Several architectures achieved validation accuracies approaching 100% during training and fine-tuning on ASVspoof2019 LA. However, their performance declined substantially when evaluated on ASVspoof2021 DF, WaveFake, and In-the-Wild. This demonstrates that strong performance on the source-domain validation data was insufficient to predict performance across different synthesis systems and recording conditions.

These results do not, by themselves, establish that the models overfitted or memorized their training samples. Instead, they demonstrate a gap between source-domain performance and cross-dataset generalization. The models may have learned useful characteristics of spoofed speech that were specific to the training distribution, or they may have relied partly on features that were less informative in the external datasets. Distinguishing these possibilities would require further analysis of training and validation curves, learned representations, and performance on additional unseen synthesis systems.

Importance of Evaluating Multiple Performance Metrics

Another important lesson was the need to consider ROC-AUC, EER, and evaluation accuracy together when assessing model performance. Accuracy measures classification decisions at a particular threshold, whereas ROC-AUC evaluates the relative ranking of bona-fide and

spoofed samples across thresholds. EER provides an additional measure of discrimination at the operating point where the false acceptance rate and false rejection rate are equal.

The experiments revealed considerable discrepancies between these metrics. For example, Swin Transformer 8H achieved a ROC-AUC of 0.7365 and an EER of 31.63% on WaveFake, despite an evaluation accuracy of approximately 50%. Similarly, on ASVspoof2021 DF, Swin Transformer 8H achieved the highest ROC-AUC of 0.8883 but an evaluation accuracy of only 39.16%. These results demonstrate that a model can retain some ranking ability while performing poorly at the decision threshold used for classification.

The findings therefore highlight the importance of examining score distributions, confusion matrices, and threshold transfer rather than interpreting accuracy in isolation. A threshold selected using ASVspoof2019 LA validation data may become less suitable when applied to a different dataset. Nevertheless, threshold mismatch does not fully explain the observed degradation because the increases in EER and decreases in ROC-AUC also indicate weaker discrimination in the external evaluations.

Architectural Performance and Generalization

The experiments did not identify a single architecture that consistently achieved the strongest results across all evaluation datasets. ResNet50 achieved the lowest EER of 6.47% on ASVspoof2019 LA, CNN–Transformer achieved the lowest EER of 20.70% on ASVspoof2021 DF, and Swin Transformer 8H achieved the lowest EER of 31.63% on WaveFake. On In-the-Wild, CNN–Transformer and ResNet18 shared the lowest reported EER of 47.36%, although their ROC-AUC values remained close to chance-level discrimination.

These differences indicate that the relative effectiveness of an architecture depends on the evaluation domain and the metric used. Strong source-domain results from a particular architecture do not necessarily translate into stronger performance on an unseen dataset. In particular, the results obtained by Swin Transformer 8H on WaveFake suggest that it retained more discriminative information for that evaluation than the other tested architectures, but its performance remained substantially weaker than on ASVspoof2019 LA.

Consequently, architecture selection should not rely exclusively on source-domain validation accuracy or a single benchmark result. Evaluating models against multiple unseen datasets and synthesis systems provides a more informative assessment of their potential robustness.

Fine-Tuning Does Not Necessarily Resolve Domain Shift

The cross-dataset results also demonstrate that fine-tuning on a source dataset does not automatically produce a detector that generalizes to different audio domains. Although the fine-tuning procedure achieved strong performance on ASVspoof2019 LA, the same models encountered substantial difficulties when evaluated on audio generated by MelGAN or collected from publicly available real-world sources.

One possible explanation is that the learned features were associated partly with the synthesis artefacts, recording conditions, or preprocessing characteristics represented in ASVspoof2019 LA. Such features may be less reliable when the detector encounters a different vocoder, compression conditions, or recording environment. This possibility is consistent with shortcut learning, in which a model relies on correlations that are useful in the training distribution but do not remain reliable under distribution shift.

However, the experiments do not establish that shortcut learning is the primary cause of the observed performance decline. Other factors, including differences in genuine and synthetic speech distributions, recording conditions, and fixed-length spectrogram preprocessing, may also contribute. In particular, the implemented preprocessing pipeline pads spectrograms after conversion to decibels, making it important to investigate whether padding and duration differences introduce unintended cues. Further controlled experiments would be necessary to determine the extent to which these factors affect model predictions.

6. Conclusion

This study presented a comparative evaluation of convolutional neural networks, transformer-based architectures, and a hybrid CNN–Transformer model for spectrogram-based audio deepfake detection. The experiments evaluated ResNet18, ResNet50, Vision Transformer (ViT), two Swin Transformer configurations, CNN–Transformer, DeiT-Hard, and DeiT-Soft using ASVspoof2019 LA as the source dataset. The investigation examined the influence of training strategy, architecture-specific fine-tuning, and evaluation methodology on detection performance, followed by cross-dataset evaluation using ASVspoof2021 DF, WaveFake LJSpeech–MelGAN, and In-the-Wild.

During the initial benchmarking phase, training all models under a fixed 100-epoch schedule changed the relative performance of the evaluated architectures. ResNet50 achieved an EER of 6.47%, narrowly outperforming CNN Transformer at 6.70%. This result highlighted the importance of training duration when comparing architectures with different optimization characteristics. However, subsequent architecture-specific fine-tuning produced different rankings, with Swin Architectures achieving the lowest reported EER of 5.86% and 5.97% on ASVspoof2019 LA. The change in rankings across experimental configurations indicates that model performance depends not only on architectural design but also on the training procedure, optimization configuration, and checkpoint selected for evaluation.

The experiments also highlighted the importance of validating model outputs and evaluation metrics before interpreting apparent performance improvements. Earlier fine-tuning runs produced exceptionally low EER values that were inconsistent with other evaluation measures, including accuracy, ROC-AUC, confusion matrices, and score distributions. Subsequent debugging involving layer-wise activation tracing and investigation of numerical and representation-related issues emphasized the need to verify the consistency of model behaviour across multiple diagnostic measures. This experience demonstrated that a single metric should not be interpreted in isolation and that unexpected improvements require further investigation before being accepted as evidence of better detection performance.

The cross-dataset evaluations revealed a substantial gap between source-domain performance and generalization to unseen data. On ASVspoof2021 DF, CNN–Transformer achieved the lowest EER of 20.70%, while Swin Transformer 8H obtained the highest ROC-AUC of 0.8883. On WaveFake, Swin Transformer 8H achieved the strongest results, with an EER of 31.63% and a ROC-AUC of 0.7365. In-the-Wild presented the greatest challenge, with CNN–Transformer and ResNet18 sharing the lowest reported EER of 47.36%; ResNet18 achieved the highest ROC-AUC of 0.5438. These results demonstrate that strong validation performance on ASVspoof2019 LA did not translate into equivalent performance across different synthesis systems and recording conditions.

The variation in architecture rankings across datasets further indicates that no single model consistently outperformed the others under every evaluation condition. Swin Transformer 8H

performed best in terms of EER on the fine-tuned ASVspoof2019 LA evaluation, CNN–Transformer performed best on ASVspoof2021 DF, and Swin Transformer 8H achieved the strongest results on WaveFake. Although the Swin Transformer variants retained relatively better discrimination on WaveFake, their performance still deteriorated considerably compared with the source-domain evaluation. This suggests that the features learned during training were not equally transferable across synthesis domains. Differences in generator characteristics, acoustic conditions, compression, and preprocessing may contribute to this degradation, although the current experiments do not isolate their individual effects.

Another important finding concerns the distinction between ranking discrimination and threshold-dependent classification. Several external evaluations exhibited substantial discrepancies between ROC-AUC, EER, and accuracy. In particular, some models retained measurable ranking ability despite evaluation accuracies close to chance. This behaviour illustrates that a decision threshold selected using source-domain validation data may not remain suitable when applied to an external dataset. Nevertheless, threshold mismatch alone cannot account for all the observed degradation, as the higher EERs and lower ROC-AUC values also indicate weaker discrimination on unfamiliar data. Reliable evaluation must therefore consider score distributions, class-label conventions, threshold transfer, and multiple complementary performance metrics.

Implications for Future Experiments

The findings suggest several important directions for further investigation. First, future experiments should examine strategies for improving generalization across unseen synthesis systems, including training with a wider range of vocoders and evaluating augmentation methods that simulate compression and other audio-domain variations. Controlled ablation studies would help establish whether these techniques improve generalization rather than merely increasing performance on the source dataset.

Second, the preprocessing pipeline should be investigated to determine whether fixed-length spectrogram construction introduces unintended cues. In particular, the effects of time truncation, padding, duration differences, and decibel-domain padding should be examined through controlled comparisons. Such experiments would help establish whether the observed performance degradation is partly influenced by preprocessing artefacts or predominantly reflects differences between the training and evaluation distributions.

Third, future work should investigate the features used by the models to distinguish bona-fide and spoofed speech. Analysis of frequency-domain characteristics, feature representations, and controlled counterfactual tests could help determine whether the detectors rely on generalizable evidence of synthetic speech or on dataset-specific correlations. This would provide a more direct evaluation of the shortcut-learning hypothesis suggested by the cross-dataset results.

Finally, score calibration and decision-threshold selection should be examined separately from model discrimination. Thresholds should be assessed using appropriate validation or calibration data rather than optimized on the final test sets. Additional experiments should also compare model performance across independent runs and checkpoints to evaluate training stability and the reproducibility of the observed results.

Overall, this study demonstrates that effective audio deepfake detection requires more than achieving high validation accuracy on a single benchmark. Although the evaluated

architectures achieved strong source-domain performance, their external results exposed substantial limitations in cross-dataset generalization. The principal contribution of the experiments is therefore not only the comparison of convolutional and transformer-based architectures, but also the identification of the gap between benchmark performance and reliable detection across unfamiliar audio domains. Addressing this gap through more diverse training data, carefully validated preprocessing, controlled optimization experiments, and generator-independent representations represents an important direction for future research.

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